Progress Report: How Foveated Vision Shapes Cognitive Maps in Navigation Agents

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I. Training Status

All agents are trained with DD-PPO [1] on Gibson-0+ (411 scenes) + MP3D train (61 scenes) in Habitat [2], sharing a 3-layer LSTM (512-d) backbone. Training targets are set to ensure convergence rather than a fixed frame budget.

Table I
TRAINING STATUS PER CONDITION (AS OF APRIL 12, 2026).

Condition	Frames	Target	Success	SPL	Status
Blind	179M	250M	93.2%	0.589	Training
Uniform	249M	400M	81.1%	0.460	Training
Foveated	154M	250M	90.2%	0.448	Training
Matched	287M	250M	82.4%	0.454	Past target
Fov-learned	—	250M	—	—	Not started

Design note: Uniform is trained to 400M (vs. 250M for others) because its ResNet-18 encoder requires more samples to converge. We match convergence level rather than frame count to ensure fair cross-condition comparison.

II. Probing Results (Phase 1 – Baseline)

Full 13-experiment probing analysis has been completed for all four conditions using their best available checkpoints (blind: ckpt.19, uniform: ckpt.33, foveated: ckpt.21, matched: ckpt.30). All probes use full 512-d hidden states (no PCA).

A. Global GPS and Compass Probes (1b)

Table II
GLOBAL LINEAR PROBE ACCURACY (RIDGE, $\alpha = 10$, EPISODE-LEVEL TRAIN/TEST SPLIT). HIGHER R^2 = BETTER SPATIAL ENCODING.

	GPS R^2	GPS MAE	Compass R^2	Compass MAE
Blind	**0.899**	0.020m	**0.878**	0.8°
Foveated	0.841	0.034m	0.715	2.1°
Matched	0.832	0.033m	0.607	1.3°
Uniform	0.614	0.035m	0.423	1.6°

Key observations:

- Blind dominates baseline probes—GPS/compass are directly in the input, so the LSTM trivially encodes them. Replicates Wijmans et al. [3].
- Among sighted conditions, **foveated ranks highest** on both GPS (0.841) and compass (0.715).
- Uniform is weakest (GPS 0.614)—the full visual stream may act as a crutch, reducing reliance on memory.

B. Distance-to-Goal and Control Tasks (1c, 1e–f)

All selectivity values are well above zero, confirming probes capture genuine spatial structure, not memorization artifacts.

Table III
DISTANCE-TO-GOAL PROBE AND SELECTIVITY (HEWITT & LIANG CONTROL).

	Dist-to-goal R^2	GPS selectivity	Compass selectivity
Blind	**0.984**	**0.911**	**0.890**
Foveated	0.953	0.880	0.751
Matched	0.977	0.871	0.701
Uniform	0.959	0.763	0.543

C. Multi-Layer Comparison (1d)

For the blind agent, layer 0 hidden state (h) achieves the highest GPS R^2 (0.958), while layer 1 cell state (c) is a close second (0.962). The top LSTM layer (layer 2, h) matches the default probe target (0.899). Spatial information is distributed across layers but concentrated in layers 0–1.

III. Probing Results (Phase 2 – H1: Compensatory Memory)

A. Path-History Probe (2c)

This probe decodes the agent's position at k steps *in the past* from the current hidden state, testing how far back spatial memory extends.

Table IV
PATH-HISTORY PROBE: R^2 FOR DECODING GPS AT LAG k FROM CURRENT HIDDEN STATE. HIGHER = LONGER WORKING MEMORY.

	lag-0	lag-1	lag-2	lag-3	lag-4	lag-5
Blind	0.899	0.905	0.867	0.748	0.536	0.536
Foveated	0.841	0.862	0.826	0.737	**0.685**	**0.629**
Matched	0.832	0.841	0.779	0.581	0.475	0.445
Uniform	0.614	0.555	0.393	0.386	0.008	−0.105

This is the strongest finding so far. The foveated agent retains spatial information over longer temporal horizons than any other condition:

- At lag-5, foveated R^2=0.629 vs. blind 0.536, matched 0.445, uniform −0.105.
- Foveated surpasses even blind at lags 4–5, despite blind having GPS directly in its input.
- Uniform memory decays rapidly and becomes uninformative by lag-5.

This is **direct evidence for H1 (compensatory memory)**: degraded peripheral vision forces the foveated agent to build a richer temporal trace of where it has been.

The foveated–matched comparison (0.629 vs. 0.445 at lag-5) isolates the effect of *spatial non-uniformity* from total information volume: it is the uneven quality of the foveated sensor, not simply having less visual information, that drives compensatory memory.

B. Visited-Region Probe (2d)

Table V
VISITED-REGION PROBE: CAN THE HIDDEN STATE PREDICT WHICH SPATIAL CELLS THE AGENT HAS VISITED?

	R^2	Binary accuracy
Blind	**0.834**	99.8%
Uniform	0.826	99.8%
Matched	0.808	99.8%
Foveated	0.801	99.8%

All conditions track visited regions near-perfectly (>99.8% binary accuracy). R^2 values are similar, suggesting this is a basic capability shared by all agents regardless of visual input.

C. Occupancy Map Reconstruction (2e)

Table VI
OCCUPANCY DECODING FROM HIDDEN STATES (20×20 GRID).

	R^2	Binary accuracy
Blind	−0.144	59.6%
Matched	−0.379	58.1%
Foveated	−0.397	55.2%
Uniform	−0.300	56.3%

Occupancy decoding is weak across all conditions (all $R^2 < 0$). None of the agents build detailed layout maps in their hidden states. This may require more episodes, a finer grid, or a non-linear decoder to detect.

D. Place Cells (5a)

Table VII
PLACE-CELL-LIKE UNITS (SPATIAL INFORMATION > 1 BIT) OUT OF 512 HIDDEN UNITS.

	Place cells (>1 bit)
Foveated	**475**/512
Blind	473/512
Uniform	473/512
Matched	461/512

Nearly all units carry spatial information across all conditions. Matched has slightly fewer place-cell-like units (461 vs. ∼473).

IV. PROBING RESULTS (PHASE 3 – H2: REPRESENTATIONAL DIVERGENCE)

A. Cross-Condition CKA (3a)

Linear CKA measures representational similarity between conditions (1.0 = identical geometry, 0.0 = unrelated). We compute CKA on the top LSTM hidden states (h_layer_2) using matched episodes across all condition pairs.

Key observations:

- All CKA values are near zero (<0.003), indicating that each condition develops a **fundamentally different representational geometry** for encoding space.

Table VIII
LINEAR CKA SIMILARITY (TOP-LAYER HIDDEN STATE, $n \approx 1{,}910$ STEPS).

	Blind	Uniform	Foveated	Matched
Blind	—	0.0023	0.0023	0.0011
Uniform		—	**0.0007**	0.0017
Foveated			—	0.0011

- Blind is the most dissimilar to all sighted conditions—consistent with its reliance on proprioceptive rather than visual features.
- The closest pair is uniform–foveated (0.0007), yet still near zero, showing that even sighted agents with different visual sampling diverge strongly.

B. Probe Transfer (3b)

We train a GPS probe on one condition's hidden states and evaluate it on another's. Self-accuracy (∼0.96–0.97) confirms probes work; cross-condition transfer tests whether spatial information is encoded in a shared format.

Table IX
PROBE-TRANSFER R^2: GPS PROBE TRAINED ON ROW, TESTED ON COLUMN.

Train \ Test	Blind	Uniform	Foveated	Matched
Blind	**0.973**	−9.72	−2.71	−15.72
Uniform	−7.26	**0.973**	−13.53	−11.47
Foveated	−11.25	−5.12	**0.963**	−10.66
Matched	−5.70	−11.74	−15.52	**0.959**

Cross-condition transfer is **catastrophically negative** in all cases, confirming that spatial information is not encoded in a shared linear subspace across conditions. Combined with near-zero CKA, this provides **strong evidence for H2**: visual input type fundamentally shapes the representational geometry used to encode space, not just the amount of spatial information stored.

V. SUMMARY OF HYPOTHESES

Table X
HYPOTHESIS STATUS BASED ON CURRENT RESULTS.

Hypothesis	Status	Key evidence
H1: Compensatory memory	**Supported**	Foveated lag-5 R^2=0.629 (best across all conditions)
H2: Representational divergence	**Supported**	All CKA <0.003; cross-transfer catastrophically negative
H3: Epistemic gaze	Not started	Learned-gaze not yet trained

VI. NEXT STEPS

1) **Complete training:** Blind (179M→250M), foveated (154M→250M), uniform (249M→400M). Re-probe with final checkpoints.
2) **Shortcut behavioral test:** Job running for blind. Will test whether persistent LSTM memory functionally aids navigation vs. reset memory.

3) **Foveated-learned (5th condition):** Train the learned-gaze agent to test H3. Requires launching a new training run.
4) **Visualization:** PCA/t-SNE of hidden states colored by position and condition for the paper.
5) **Re-probe after convergence:** Once all training finishes, re-run full probing suite on final checkpoints for the definitive comparison.

References

[1] E. Wijmans, A. Kadian, A. Morcos, S. Lee, I. Essa, D. Parikh, D. Batra, and M. Savva, "DD-PPO: Learning near-perfect pointgoal navigators from 2.5 billion frames," in *International Conference on Learning Representations (ICLR)*, 2020.

[2] M. Savva, A. Kadian, O. Maksymets, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, V. Koltun, J. Malik, D. Batra, and D. Parikh, "Habitat: A platform for embodied AI research," in *International Conference on Computer Vision (ICCV)*, 2019.

[3] E. Wijmans, M. Savva, I. Essa, S. Lee, A. Morcos, and D. Batra, "Emergence of maps in the memories of blind navigation agents," in *International Conference on Learning Representations (ICLR)*, 2023.